

Biological Networks with Singular Jacobians: Their Origins and Adaptation Criteria

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- Systems respond to a stimulus, then return to baseline
- Important concept: **Robust Perfect Adaptation (RPA)**
 - Output returns to the same steady-state value
 - Independent of input strength

Limitations of Existing Theory

- Biological networks are modeled using **ODE systems**
- Behavior near steady state analyzed via the **Jacobian matrix**
- Existing theory assumes:
 - **Non-singular Jacobian** (invertible)
 - **Hyperbolic equilibrium**

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 - Causes:
 - Conservation laws
 - Complex molecular interactions

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- **Problem:**
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 - Causes:
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 - Complex molecular interactions
- These systems are **not covered by existing theory**

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 - Applies to:
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- **Takeaway:** Extends adaptation theory to realistic biological models

Mathematical Notation: State Space

We consider a system with state space

$$\mathcal{P} \subset \mathbb{R}^N.$$

The state of the system at time t is

$$\mathbf{P}(t) = [P_1(t), \dots, P_N(t)]^T \in \mathcal{P},$$

where each component $P_i(t)$ represents a node of the system.

Examples of nodes include:

- proteins,
- genes
- RNA transcripts,

Mathematical Notation: Input Space

Let

$$\mathcal{U} \subset \mathbb{R}^M$$

be the input space, and

$$\mathbf{U}(t) = [U_1(t), U_2(t), \dots, U_M(t)]^T \in \mathcal{U}$$

be the time-dependent input to the system.

For the system, inputs represent quantities that can be externally manipulated, such as:

- Applying a drug to the system and increasing the dosage over time,
- perturbations to the system coefficients.

Mathematical Notation: Output

The output space will be

$$Y \subset \mathbb{R},$$

and

$$h : \mathcal{P} \rightarrow Y, \quad h(\mathbf{P}) = \sum_{i=1}^N h_i P_i$$

is a linear function of the state variables.

By choice h_i , we can define the output to be a single state variable, or a weighted sum of several variables in the system. Therefore, let

$$h = [h_1, \dots, h_N] \quad \text{with} \quad 0 \leq h_i \leq 1, \forall i$$

be the (row)vector of coefficients which determine the weighted contribution from each node to the total output of the system.

Mathematical Notation: System Dynamics

Let

$$\mathcal{F} : \mathcal{P} \times \mathcal{U} \rightarrow \mathbb{R}^N$$

be a continuously differentiable function describing the system dynamics.

The dynamical system is

$$\frac{dP_i}{dt} = f_i(\mathbf{U}, \mathbf{P}), \quad i = 1, \dots, N.$$

In matrix form, the nonlinear dynamical system $\frac{dP_i}{dt} = f_i(\mathbf{U}, \mathbf{P})$ can be written as

$$\mathbf{P}' = \mathbf{F}(\mathbf{P}, \mathbf{U}).$$

Mathematical Notation: Jacobian and Steady State

The Jacobian matrix with respect to the state variables is

$$\mathbf{J}(\mathbf{P}, \mathbf{U}) = D_{\mathbf{P}}F(\mathbf{P}, \mathbf{U}).$$

At a steady state $\mathbf{P}^* = g(\mathbf{U})$, the system satisfies

$$f_i(\mathbf{P}^*, \mathbf{U}) = 0, \quad i = 1, \dots, N$$

The linearization of the system $\frac{dP_i}{dt} = f_i(\mathbf{U}, \mathbf{P})$ at the steady state is given by

$$\mathbf{P}' = D_{\mathbf{P}}F(\mathbf{P}^*, \mathbf{U}).$$

Origins of Singularities

We study the conditions under which a singular Jacobian may arise. For simplicity:

- We suppress the dependence on the input $\mathbf{U}$,
- focuses on the system at steady state.

The results remain valid when the full input-output structure is considered.

A singular Jacobian may arise due to

- Conservation laws

The following result characterizes the Jacobian in this situation.

Conservation laws

Proposition 1. Consider a biological system with $N \geq 2$ nodes which contains a linear conservation law. The system is defined as follows

$$\frac{d\mathbf{P}}{dt} = \mathbf{F}(\mathbf{P}), \quad \mathbf{P}_{\text{tot}} = \mathbf{c}^T \mathbf{P}(\mathbf{t})$$

where $\mathbf{c} \neq 0$ is a vector of coefficients in the conservation law and $\mathbf{P}_{\text{tot}}$ is the total concentration which remains constant. Then,

$$\det(\mathbf{J}(\mathbf{P}(\mathbf{t}))) = 0, \quad \forall t \geq 0.$$

Idea of Proof..

- Differentiate the conservation law in time.
- Use the system equations to relate the derivatives.
- Express one equation as a combination of the others. This creates linear dependence between rows of the Jacobian
- The Jacobian is singular and $\det(J) = 0$.

Conservation laws

Proposition 2. Consider a biological system with $N \geq 2$ nodes which contains a simple non-linear conservation law. The system is defined as follows:

$$\frac{d\mathbf{P}}{dt} = \mathbf{F}(\mathbf{P}), \quad \mathbf{P}_{\text{tot}} = \sum_{i=1}^N v_i(\mathbf{P}_i)$$

where $\mathbf{v} = [v_1(P_1) \cdots v_n(P_N)] \neq 0$, $v_i \in \mathbf{C}^2(\mathcal{P})$, and for

$$v_i(\mathbf{P}_i) \neq 0, \quad \frac{dv_i}{d\mathbf{P}_i} \neq 0, \quad \text{and} \quad \frac{dv_i(\mathbf{P}_i^*)}{d\mathbf{P}_i} \neq 0.$$

$\mathbf{P}_{\text{tot}}$ is the total concentration which remains constant. Then,

$$\det(\mathbf{J}(\mathbf{P}^*)) = 0.$$

Note: If we allow $\frac{dv_i}{d\mathbf{P}_i} = 0$, this means that $v_i(\mathbf{P}_i) = b$, where b is a constant. However, this would be independent of $\mathbf{P}_i$.

Conservation laws

Proposition 3. Consider a biological system with $N \geq 2$ nodes which contains a nonlinear conservation law. Let

$$\mathbf{M} = \{i : \mathbf{P}_i \text{ appears in the conservation law}\}, \quad 2 \leq m = |\mathbf{M}| \leq N.$$

The system is defined as follows:

$$\frac{d\mathbf{P}}{dt} = \mathbf{F}(\mathbf{P}), \quad \mathbf{P}_{\text{tot}} = \mathbf{v}(\mathbf{P}_{i_1}, \dots, \mathbf{P}_{i_m}), \quad i_j \in \mathbf{M} \text{ are distinct}$$

where $v \in \mathbf{C}^2(\mathcal{P}^m)$,

$$\frac{\partial v}{\partial \mathbf{P}_{i_j}}(\mathbf{P}_{i_1}^*, \dots, \mathbf{P}_{i_m}^*) \neq 0$$

for non-trivial steady state $\mathbf{P}^*$, and $\mathbf{P}_{\text{tot}}$ is the total concentration which remains constant. Then,

$$\det(\mathbf{J}(\mathbf{P}^*)) = 0$$

for this system.

Conservation law example

Consider the oxidation catalysis model considered in Emelianenko et al.,

$$\begin{cases} \dot{x} = -k_1x + k_2yz \\ \dot{y} = k_1x - k_2yz - k_3y \\ \dot{z} = -k_2yz \end{cases}$$

By combining the system equations,

$$\frac{d}{dt} \left[x + y - \frac{k_3}{k_2} \ln(z) \right] = 0$$

The system has nontrivial steady state solution $x^* = y^* = 0$, $z^* \neq 0$.

$$\mathbf{J}^* = \begin{bmatrix} -k_1 & k_2z^* & 0 \\ k_1 & -(k_2z^* + k_3) & 0 \\ 0 & -k_2z^* & 0 \end{bmatrix}$$

which clearly has $\det(\mathbf{J}^*) = 0$.

Special network structures

Singular Jacobians may arise not only from conservation laws.

- **Special network structures** (a multi-node complex.)

Proposition 4. Consider an N -node network, $N \geq 3$, in which one node, say $\mathbf{P}_N$, is a function of the other nodes within the network. Let the system be defined as:

$$\begin{cases} \frac{d\mathbf{P}}{dt} = \mathbf{F}(\mathbf{P}) \\ \mathbf{P}_N = \mathbf{v}(\mathbf{P}_1, \dots, \mathbf{P}_{N-1}) \end{cases}$$

where $\mathbf{v} \in \mathbf{C}^2(\mathcal{P}^{N-1})$ and at least one

$$\frac{d\mathbf{P}_N}{d\mathbf{P}_i} \neq 0 \quad \text{for } i = 1, \dots, N-1.$$

Then,

$$\det(\mathbf{J}(\mathbf{P}^*)) = 0$$

for this system.

Adaptation in Biological Networks

Goal: Examine how singular Jacobians affect adaptation criteria

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Two cases to consider:

- **Case 1:** $\det(D_{\mathbf{p}}\mathbf{F}) \neq 0$ — Non-singular
- **Case 2:** $\det(D_{\mathbf{p}}\mathbf{F}) = 0$ — Singular

Adaptation in Biological Networks

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Plan:



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OPEN

The topological requirements for robust perfect adaptation in networks of any size

Robyn P. Araujo^{1,2} & Lance A. Liotta³

Adaptation in Biological Networks

- **Case 1: Non-Singular Systems (Known Results)**

Assumption: $\det(D_{\mathbf{p}}\mathbf{F}) \neq 0$ (invertible Jacobian)

Multiple-input/multiple-output case:

Adaptation occurs if:

$$hC^T D_{\mathbf{u}}\mathbf{F} = 0$$

- $h = [h_1, \dots, h_N]$: output coefficient vector
- C : cofactor matrix of $D_{\mathbf{p}}\mathbf{F}$

Adaptation in Biological Networks

Nonlinear dynamical system:

$$\mathbf{P}' = \mathbf{F}(\mathbf{P}, \mathbf{U}) = \mathbf{F}(\mathbf{P}(\mathbf{U}), \mathbf{U})$$

Full derivative with respect to input:

$$\frac{d\mathbf{F}}{d\mathbf{U}} = D_{\mathbf{P}}\mathbf{F} \cdot D_{\mathbf{U}}\mathbf{P} + D_{\mathbf{U}}\mathbf{F}$$

Evaluated at steady state $\mathbf{P}^* = g(\mathbf{U})$:

$$D_{\mathbf{P}}\mathbf{F}(g(\mathbf{U}), \mathbf{U}) \cdot D_{\mathbf{U}}\mathbf{P}(g(\mathbf{U}), \mathbf{U}) = -D_{\mathbf{U}}\mathbf{F}(g(\mathbf{U}), \mathbf{U})$$

Adaptation in Biological Networks

Since $D_{\mathbf{P}}\mathbf{F}$ is invertible:

$$(D_{\mathbf{P}}\mathbf{F})^{-1} = \frac{\text{adj}(D_{\mathbf{P}}\mathbf{F})}{\det(D_{\mathbf{P}}\mathbf{F})} = \frac{C^T}{\det(D_{\mathbf{P}}\mathbf{F})}$$

From $D_{\mathbf{P}}\mathbf{F}(g(\mathbf{U}), \mathbf{U}) \cdot D_{\mathbf{U}}\mathbf{P}(g(\mathbf{U}), \mathbf{U}) = -D_{\mathbf{U}}\mathbf{F}(g(\mathbf{U}), \mathbf{U})$

$$D_{\mathbf{U}}\mathbf{P} = -(D_{\mathbf{P}}\mathbf{F})^{-1}D_{\mathbf{U}}\mathbf{F} = -\frac{C^T}{\det(D_{\mathbf{P}}\mathbf{F})}D_{\mathbf{U}}\mathbf{F}$$

Multiply by h :

$$\det(D_{\mathbf{P}}\mathbf{F}) hD_{\mathbf{U}}\mathbf{P} = -hC^T D_{\mathbf{U}}\mathbf{F}$$

Adaptation requires $hD_{\mathbf{U}}\mathbf{P} = 0 \rightarrow$ yields condition $hC^T D_{\mathbf{U}}\mathbf{F} = 0$.

Transition to Case 2 — Singular Systems

What happens when $\det(D_{\mathbf{p}}\mathbf{F}) = 0$?

- Standard approach fails — Jacobian is **not invertible**
- Cannot use cofactor-based method directly
- Need a **generalized adaptation criterion**

Generalized Adaptation Criterion for Singular Systems

- **Case 2: Singular Systems ($\det(D_P F) = 0$)**

Challenge:

- Jacobian is not invertible
- Cannot use $(D_P \mathbf{F})^{-1}$ or cofactor method

Solution:

- Use Singular Value Decomposition (SVD)
- Derive generalized condition using Moore-Penrose pseudoinverse

Generalized Adaptation Criterion for Singular Systems

Singular Value Decomposition (SVD)

For any matrix $A \in \mathbb{R}^{N \times N}$ with rank $r < N$:

$$D_{\mathbf{p}}\mathbf{F} = W\Sigma V^T$$

Reduced form (keeping only non-zero singular values):

$$D_{\mathbf{p}}\mathbf{F} = W_1\Sigma_1 V_1^T$$

- $\Sigma_1 \in \mathbb{R}^{r \times r}$: diagonal matrix of non-zero singular values
- $W_1 \in \mathbb{R}^{N \times r}$: left singular vectors (non-zero singular values)
- $V_1 \in \mathbb{R}^{N \times r}$: right singular vectors (non-zero singular values)

Generalized Adaptation Criterion for Singular Systems

Moore-Penrose Pseudoinverse

Pseudoinverse of $D_{\mathbf{p}}\mathbf{F}$:

$$(D_{\mathbf{p}}\mathbf{F})^{\dagger} = V_1 \Sigma_1^{-1} W_1^T$$

Properties:

- Exists for any matrix (singular or non-singular)
- Provides the minimum norm least squares solution to linear systems

Generalized Adaptation Criterion for Singular Systems

Starting from steady-state equation

$$D_P \mathbf{F} \cdot D_U \mathbf{P} = -D_U \mathbf{F}$$

$$D_U \mathbf{P} = -(D_P \mathbf{F})^{-1} D_U \mathbf{F}$$

Apply pseudoinverse:

$$D_U \mathbf{P} = -(D_P \mathbf{F})^\dagger D_U \mathbf{F} = -V_1 \Sigma_1^{-1} W_1^T D_U \mathbf{F}$$

Multiply by output vector h :

$$h D_U \mathbf{P} = -h V_1 \Sigma_1^{-1} W_1^T D_U \mathbf{F}$$

Generalized Adaptation Criterion for Singular Systems

Proposition 5

A biochemical system $\frac{dP_i}{dt} = f_i(U, P), i = 1, \dots, N$ with inputs $U \in \mathbb{R}^M$ for which $\det(D_P \mathbf{F})(P^*) = 0$ exhibits adaptation if

$$hV_1\Sigma_1^{-1}W_1^T D_U \mathbf{F} = 0$$

at the system's steady state P^* .

Where:

- V_1, Σ_1, W_1 from SVD of $D_P \mathbf{F}$
- h : output coefficient vector
- $D_U \mathbf{F}$: input sensitivity matrix

Generalized Adaptation Criterion for Singular Systems

Special case: Single input, single output

- $M = 1$
- $h = e_O^T$ (output node O)
- $D_U \mathbf{F} = \frac{\partial f_I}{\partial I} e_I$ (input applied to node I)

Corollary

A biochemical system $\frac{dP_i}{dt} = f_i(U, P)$ for which $\det(D_P \mathbf{F})(P^*) = 0$ exhibits RPA if, at steady state,

$$e_O^T V_1 \Sigma_1^{-1} W_1^T e_I = 0$$

where V_1, W_1, Σ_1 are obtained from the SVD of $D_P \mathbf{F}$.

Conclusion

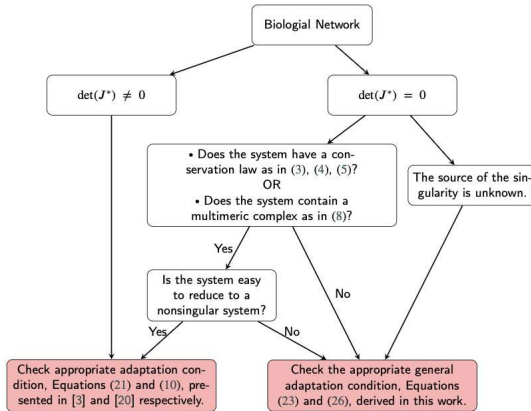


Figure 1: Flowchart detailing process of checking adaptation.

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